

WIFIND: Enhancing Wi-Fi Fingerprint Indoor Localization With a Spatially Conditioned Diffusion Model-Based Data Augmentation

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Abstract—Fingerprint-based indoor positioning systems require exhaustive site-surveying processes to establish the radio maps of wireless signals, such as Wi-Fi, at the target sites for localization, which are extremely time-consuming. Unfortunately, a rough, coarse-grained site-surveying run gathers inadequate radio fingerprints to produce lower-resolution radio maps. To address such a time-versus-accuracy dilemma, this paper proposes WIFIND, a novel approach employing diffusion models to generate additional details to reinforce the collected fingerprints. WIFIND employs a diffusion model trained on real-world fingerprint data to generate synthetic Channel State Information (CSI) that mimics actual Wi-Fi signals. Using a Spatially Conditioned Diffusion Model (SCDM), SCDM generates synthetic CSI data based on the given position, ensuring the data accurately reflects the spatial variations of Wi-Fi signals within the environment. This results in a richer and more diverse dataset, enhancing the model's ability to localize devices precisely. Extensive experiments demonstrate that WIFIND not only improves upon the baseline models trained with actual data but also outperforms existing state-of-the-art methods for CSI data augmentation. Preliminary results on the Ultra Dense Indoor MaMIMO dataset show that WIFIND achieves improvements ranging from 47.56% to 78.53% compared to the baseline, respectively.

Index Terms—Indoor localization, Data augmentation, Diffusion model, Channel state information.

I. INTRODUCTION

In recent years, Indoor Positioning Systems (IPS) have become essential for many critical applications, including asset tracking, inventory management, emergency response coordination, and personalized advertising [1]–[3]. Among the technologies that enable IPS, Wi-Fi fingerprinting [4], [5] stands out due to its ability to utilize existing Wi-Fi infrastructure without additional installations. This methodology effectively maps signal strengths from various access points into detailed location “fingerprints,” offering precision and efficiency in indoor navigation.

However, despite its benefits, Wi-Fi fingerprinting faces notable challenges in data collection. These challenges include the labor-intensive nature of the process, which typically requires over a hundred person-hours even in moderately sized facilities, with these demands escalating significantly in larger spaces. The accuracy of the data is also critically vulnerable to fluctuations in environmental conditions and potential inconsistencies in hardware performance, which can severely compromise the system's efficiency and scalability.

These challenges pose a substantial barrier to the widespread adoption and practical deployment of Wi-Fi fingerprinting.

To overcome the above hurdles, we propose WIFIND, an integrated, systematic Wi-Fi fingerprint positioning approach that incorporates diffusion-based data augmentation. The main contributions of WIFIND are as follows:

- 1) **Works on Channel-State Information data:** Unlike previous approaches that simply augmented RSSI (Received Signal Strength Indicator) data [6]–[11], WIFIND can generate adequate pseudo-CSI data to augment the already information-rich CSI fingerprints collected exhaustively in the real world.
- 2) **Simultaneous Label and Data Generation:** WIFIND employs a conditional diffusion model to generate synthetic CSI data based on given positional information, eliminating the need for separate pseudo-labeling [8], [9]. This approach ensures the simultaneous generation of both position labels and CSI data, enabling more accurate and spatially consistent augmentation.
- 3) **Enhanced Positioning Accuracy:** Preliminary experimental results on publicly available CSI datasets show that WIFIND successfully reduced positioning errors by up to 78.53%, outperforming existing state-of-the-art methods for CSI data augmentation.

The rest of this paper is organized as follows: Section II reviews related work on IPS and data augmentation. Section III details our methodology for data augmentation using diffusion models and CNNs. Section IV outlines the model settings and data preparation. Section V presents experimental results and discusses the impact of the augmentation methods. Finally, the main conclusions and future implications of this study are summarized in Section VI.

II. RELATED WORKS

A. Wi-Fi Fingerprint Indoor Positioning

Indoor positioning systems (IPS) are categorized into two methods: ranging-based and scene analysis. Technologies such as Ultra-Wideband (UWB) [12] offer high precision, while Bluetooth (BLE) [13] is commonly used for proximity estimation. Radio Frequency Identification (RFID) [14] is effective for object tracking, and Radar [15] leverages signal reflection for detection. Among the scene analysis methods,

Wi-Fi fingerprinting [4], [5] is particularly noteworthy due to its utilization of existing Wi-Fi infrastructure, eliminating the need for additional hardware.

In the evolution of Wi-Fi signals for fingerprint positioning and other classification tasks, the shift from RSSI to CSI [16]–[20] has improved the precision of signal analyses and the robustness of the techniques. CSI provides hundreds of detailed information about the signal's properties, such as phase and amplitude for each subcarrier, unlike RSSI, which solely offers a single signal strength measurement. Simultaneously, the field has advanced from traditional methodologies like K-Nearest Neighbors (KNN) [21] to adopting modern machine learning approaches. The integration of convolutional neural networks (CNNs) [22]–[25] has notably enhanced the performance and reliability of IPS, reflecting strides in computational methodologies for signal analysis.

B. Data Augmentation

The labor-intensive nature of collecting Wi-Fi fingerprints poses a substantial barrier to scalability and deployment. To address this, data augmentation has been essential. Early methods used DNNs and CNNs to enrich datasets [6], [7], while more advanced generative models like Variational Autoencoders (VAE) [11], [26] and Generative Adversarial Networks (GANs) [9], [27] have enabled synthetic data generation for RSSI-based augmentation. However, augmenting CSI data is more complex due to its sensitivity to environmental changes. Some studies have used intrinsic CSI characteristics like phase and amplitude [16], but more sophisticated techniques are needed to capture its intricate nature fully.

Among generative models, diffusion models have recently demonstrated superior performance [28]–[30], producing high-quality synthetic data across diverse applications, including image, audio, and video synthesis [31]–[33]. Given their proven effectiveness, diffusion models were selected to enhance CSI data in our indoor positioning study. This choice is motivated by the potential of diffusion models to overcome challenges in data collection [34], [35]. Additionally, diffusion models have been shown to apply innovative techniques to identification and classification tasks on Wi-Fi data [36]–[38], enhancing the reliability of applications.

III. METHODOLOGY

This section presents WIFIND's methodology, which utilizes a Spatially Conditioned Diffusion Model (SCDM) to generate synthetic CSI data. Fig. 1 outlines the step-by-step process from actual data collection to the generation of augmented datasets and their application in positioning.

The core idea is to enrich real-world CSI data by generating realistic and spatially consistent synthetic data conditioned on precise 2D positional information (x, y). Unlike other generative methods [27], [38] that rely on location IDs, WIFIND conditions the diffusion model directly on positional coordinates, allowing the model to capture detailed spatial relationships within the environment. This approach enables

the generation of synthetic data that accurately fills the gaps between the actual measurement points.

The workflow begins with utilizing actual CSI data to train the SCDM. The trained SCDM generates synthetic CSI samples that align with the spatial characteristics of the environment, including data interpolated between actual data points. These enriched data sets are then used to train a CNN-based positioning model, adapted from [25], for localization with improved accuracy.

A. Diffusion Process with Positional Conditioning

The SCDM employed in this study is designed to generate synthetic CSI data that closely mimics real-world data in statistical properties while conditioned on positional information.

The diffusion model consists of a forward process and a reverse process. In the forward process, Gaussian noise is progressively added to the original data over several steps. Such a process can be formulated as follows:

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(0, \mathbf{I}) \quad (1)$$

where $\mathbf{x}_t$ represents the data at time step t , α_t is a variance schedule, and $\mathbf{z}_t$ is Gaussian noise.

In the reverse process, the model learns to reconstruct the original data by iterative denoising the results from the forward process. Such a generative process, conditioned on positional information, ensures the synthetic CSI data aligns with specific spatial characteristics:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(\mathbf{x}_t, \text{position}) \right) \quad (2)$$

where $\epsilon_\theta(\mathbf{x}_t, \text{position})$ represents the noise estimation at step t , conditioned on the given positional embedding.

B. Position Embedding for Conditional CSI Generation

Our SCDM generates synthetic CSI data based on 2D spatial positions (x,y) rather than class labels. This approach allows our model to provide finer control over the generated data by conditioning it on exact spatial coordinates. We use a linear projection layer to expand the 2D positional inputs into a 128-dimensional embedding space, enabling the model to learn the complex relationships with the CSI data.

Once projected, the position embeddings are integrated into the diffusion process via cross-attention mechanisms, ensuring the generated CSI data remains consistent with the given position. This approach allows the model to generate synthetic CSI data that reflects spatial variation, particularly for locations between actual measurement points. By learning these positional relationships, the model effectively augments the dataset, improving the realism of the generated data.

C. Architecture of the Spatially Conditioned Diffusion Model

Our SCDM architecture, shown in Fig. 2, follows the UNet design [39] and consists of four DownBlocks, one MidBlock, and four UpBlocks. Each block incorporates two ResNet layers with SiLU activation functions and group normalization.

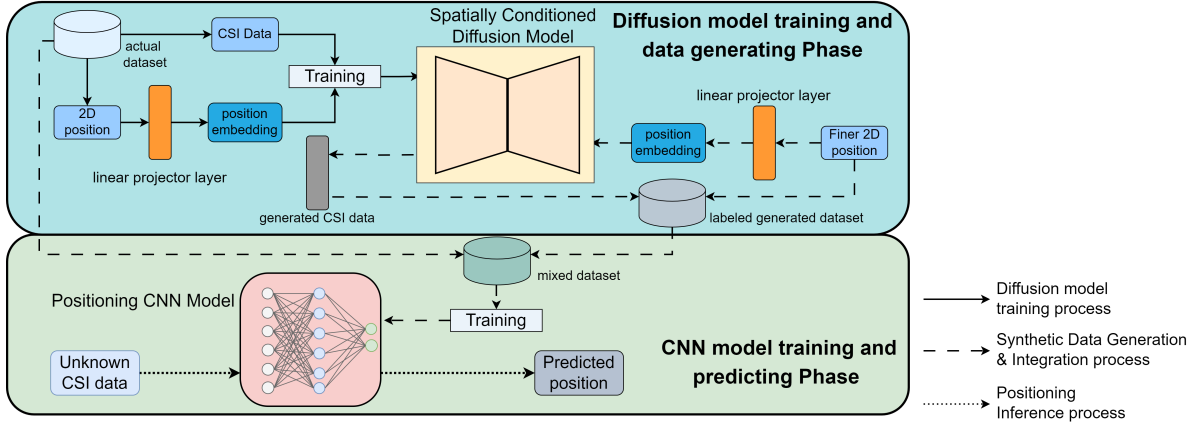


Fig. 1. Work flow of WIFIND.

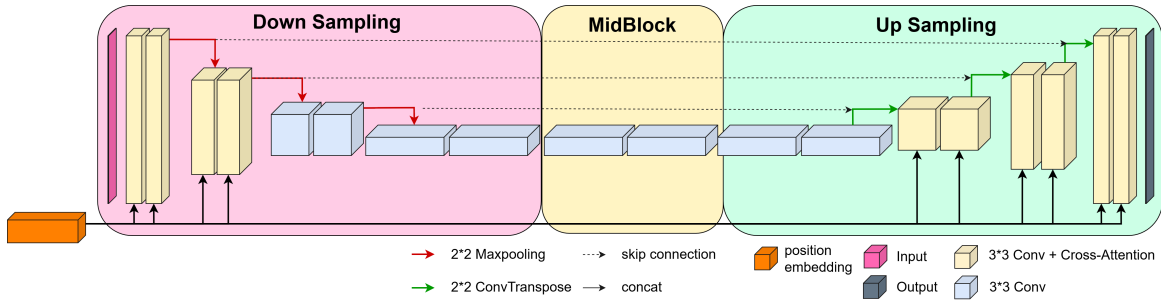


Fig. 2. Unet architecture of the SCDM.

In the downsampling phase (the leftmost portion in Fig. 2), max pooling is used to reduce spatial dimensions. The first two DownBlocks use cross-attention to incorporate positional embeddings, ensuring spatial consistency. The last two DownBlocks reduce spatial dimensions without cross-attention, further compressing the data. The MidBlock (the center portion in Fig. 2) refines the compressed feature maps at the lowest resolution before the upsampling phase. In the upsampling phase (the rightmost portion in Fig. 2), the ConvTranspose layer restores spatial dimensions. The last three UpBlocks use cross-attention to maintain spatial accuracy and align the generated data with the positional embeddings.

IV. EXPERIMENT SETUP

To validate the methodology proposed in the previous section, a WIFIND prototype was constructed as the reference platform for data augmentation enhancement on real-world CSI fingerprint datasets. The WIFIND testbed was developed in Python 3 and executed on Google Colaboratory (Colab) [40] with NVIDIA A100 GPU acceleration. Details of the experimental setup, including the datasets, configuration parameters, and data preprocessing, are described below.

A. Datasets

Our study utilizes the Ultra Dense Indoor MaMIMO dataset [41], collected using the 64-antenna KU Leuven Massive MIMO testbed. This dataset features thousands of CSI samples

from four antenna array topologies—URA LoS (Uniform Rectangular Array, Line of Sight), URA NLoS (Non-Line of Sight), ULA (Uniform Linear Array), and DIS (Distributed). These topologies are designed to simulate diverse indoor environments and signal propagation behaviors, comprehensively representing real-world scenarios.

The original dataset of each topology consisted of 252,004 samples collected in four distinct areas, each mapped on a 251×251 grid with positions recorded at 5-millimeter intervals. Each data sample is a 64×100 matrix representing the CSI at each location, with 64 corresponding to the number of BS antennas and 100 to the number of subcarriers.

To assess model performance with reduced data density, we created a sparse subset by sampling every 50 millimeters from the original dataset, resulting in 2,704 samples on a 26×26 grid. This reduced dataset provides a significantly lower density of actual data points, which we use to train our diffusion model. The model then generates 10,404 synthetic samples to match the denser sampling grid of every 25 millimeters (51×51 grid). These synthetic samples include interpolations between the original real measurements, effectively filling in the gaps between the sparse data points.

B. Configurations

The SCDM and CNN positioning models were trained for 200 epochs with a 4:1 train-to-validation data split. The SCDM

used a batch size of 32 and a maximum time step (T) of 500, while the positioning CNN also employed a batch size of 32. Table I details the configuration parameters of our experiment and other related settings.

C. Preprocessing

1) *For SCDM*: To fit the network architecture (Fig. 2), the original 64×100 CSI data is padded to 64×104 by adding two zero columns on each side, ensuring compatibility with the convolutional blocks. The padded data $64 \times 104 \times 2$ is processed through convolutional layers, initially expanding to $64 \times 104 \times 160$ in the first block. Subsequent layers reduce the spatial dimensions while increasing feature depth: $32 \times 52 \times 320$, then $16 \times 26 \times 640$, and finally $8 \times 13 \times 1280$. The MidBlock processes these compressed features to preserve key characteristics before the UpBlocks restore the original spatial dimensions, ultimately returning to $64 \times 104 \times 2$.

2) *For Positioning*: The CSI data underwent several preprocessing steps to expand the feature set and enhance its utility for the positioning model. These preprocessing steps were applied across all datasets to ensure consistency and maximize their effectiveness.

- **Conversion to Polar Domain**: The complex-valued CSI data were transformed into polar form to extract the amplitude and phase components.
- **Inverse Fast Fourier Transformation (IFFT)**: This transformation shifted the data from the frequency domain to the time domain, extracting two time-domain features.
- **Feature Concatenation**: Finally, the raw data and the new polar and time-domain features were merged into a comprehensive matrix $I \in \mathbb{R}^{64 \times 100 \times 6}$ to accommodate all 6 total features.

V. RESULTS AND DISCUSSIONS

In this section, we present and analyze the results of our experiments. We evaluate the models on data subsets to assess how the proposed augmentation method consistently improves positioning accuracy. We also discuss WIFIND's ability to mitigate overfitting, analyze training, and generate time for augmented datasets.

A. Performance of positioning with different augmentation

We compared the performance of WIFIND with several state-of-the-art methods, including Phase Augmentation, Amplitude Augmentation [16], and a Pseudo-labeling method.

TABLE I
TRAINING CONFIGURATION FOR MODELS

Parameter	Value (SCDM)	Value (CNN)
Epochs	200	200
Batch Size	32	32
Learning rate	0.0005	0.001
Time Step T	500	–
Noise schedule	cosine	–
Input Size	$64 \times 104 \times 2$	$64 \times 100 \times 6$

The Pseudo-labeling method generates synthetic labels based on predictions from a CNN trained solely on 2,704 actual data samples. It uses the same pseudo data generated by the diffusion model but without the positional conditioning that WIFIND employs. For Phase and Amplitude Augmentation, models were trained on 2,704 actual samples and $2,704 \times 4$ (10,816) augmented data points. The WIFIND method was trained on 2,704 actual data points and 10,404 pseudo data points generated with finer positional conditions.

Table II summarizes each method's mean positioning errors (in millimeters) across four scenarios: URA LoS, URA NLoS, ULA LoS, and DIS LoS. Meanwhile, Fig. 3 and Fig. 4 show the cumulative distribution function (CDF) of positioning errors for each method in the LoS and NLoS environments.

According to Table II, Fig. 3, and Fig. 4, WIFIND consistently outperformed other augmentation methods across all test scenarios, achieving superior positioning accuracies. Particularly in Line-of-Sight (LoS) scenarios, WIFIND achieved the lowest errors across all environments, excelling in the more complex and challenging NLoS conditions.

The NLoS environment, characterized by signal obstructions and increased difficulty, showcased WIFIND's significant advantage over other methods, including Phase and Amplitude Augmentations. By generating synthetic data conditioned on positional information, WIFIND effectively captured more signal propagation details than simple augmentations to phases and/or amplitudes. This ability to produce higher-quality, realistic CSI data led to WIFIND's substantial improvements in positioning accuracy, reducing the average error by as much as 78.53% compared to the baseline in NLoS conditions.

Table II also demonstrates a comparison between using the same number of actual data points (10,404) and generated data points. Remarkably, the generated data even outperforms the actual data. In LoS conditions, WIFIND reduces errors from 107.87 mm to 62.55 mm, achieving a 42.01% reduction. Similarly, in NLoS conditions, WIFIND reduces errors from 351.79 mm to 206.84 mm, representing a 41.20% reduction.

B. Training Dynamics and Overfitting

Fig. 5 and Fig. 6 compare the training and validation curves for the No Augmentation baseline and the WIFIND method. The No Augmentation approach shows clear signs of overfitting, with the validation loss significantly higher than the training loss throughout the learning process. This gap indicates that the model struggles to generalize to unseen data, particularly in more challenging NLoS environments.

The WIFIND method demonstrates superior generalization. Its training and validation curves are closely aligned, indicating that the synthetic data generated by WIFIND effectively mitigates overfitting. By enriching the dataset with realistic and diverse samples conditioned on positional data, WIFIND reduces the gap between training and validation losses, particularly in complex NLoS scenarios where signal variations are more pronounced. This ability to generate high-quality synthetic data enhances model generalization, as reflected in the reduced validation loss.

TABLE II
MEAN POSITIONING ERRORS (UNIT: MILLIMETERS)

Method	URA_LoS	ULA_LoS	DIS_LoS	Average of LoS	URA_NLoS
No aug (2704 data)	175.1480	173.4047	135.9165	161.4897	963.2425
Pseudo-labeling	134.6590	138.9529	109.3935	127.6685 (↓ 20.96%)	888.5391 (↓ 7.75%)
Phase aug [16]	76.5525	51.9587	79.3006	69.2706 (↓ 57.11%)	496.9993 (↓ 48.43%)
Amplitude aug [16]	83.1707	92.3223	81.9485	85.8138 (↓ 46.88%)	711.8108 (↓ 26.08%)
WIFIND (Our)	75.3928	40.9987	71.2680	62.5532 (↓ 61.28%)	206.8441 (↓ 78.53%)
No aug (10404 data)	83.5334	121.0286	119.0370	107.8663	351.7927

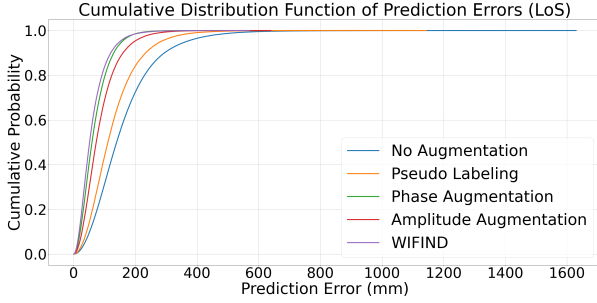


Fig. 3. CDF of positioning errors in three LoS datasets.

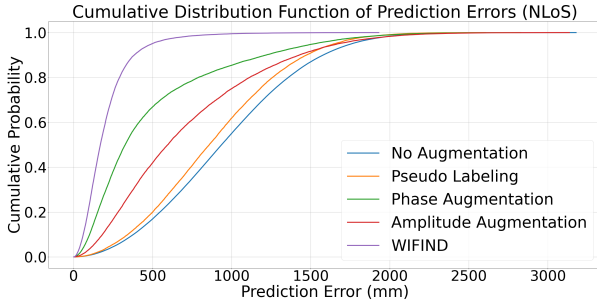


Fig. 4. CDF of positioning errors in the URA NLoS dataset.

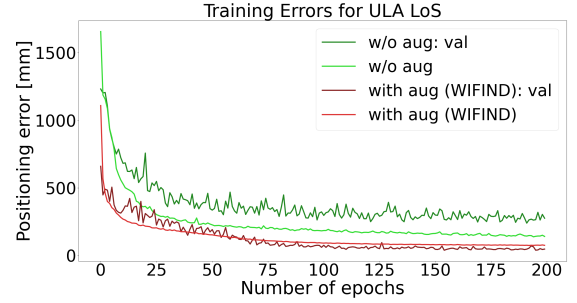


Fig. 5. Training curve in the ULA LoS dataset.

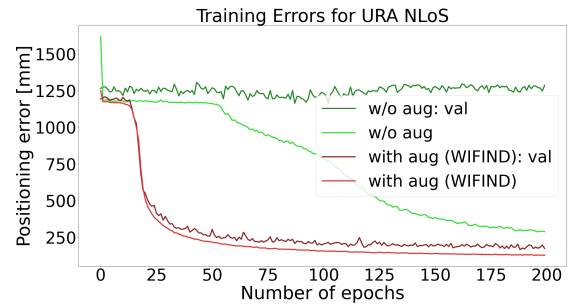


Fig. 6. Training curve in the URA NLoS dataset.

C. Discussions on Efficiency

Training the diffusion model for WIFIND takes approximately 10 hours per area, with an additional 3 hours per area required to generate a synthetic dataset. With four regions in total, the training and data generation process requires 52 hours to produce a combined synthetic dataset of 10,404 samples.

Although this process requires computational resources, it is far more time-efficient than traditional site-surveying, which can take hundreds of person-hours to collect 10,000 high-density CSI samples. With advancements in GPU technology, training and generating synthetic data efficiency continue to improve, making WIFIND even more practical.

VI. CONCLUSION AND FUTURE WORKS

Our experiment results showed that the WIFIND mechanism improves localization accuracy through data augmentation by up to 78.53% over baseline methods across all tested scenarios. Attribute to the generated synthetic data with actual and pseudo-CSI fingerprints plus finer positional conditions,

WIFIND consistently achieved lower positioning errors, most notably in NLOS scenarios. Leveraging the Spatially Conditioned Diffusion Model (SCDM), WIFIND not only reduces the labor involved in fingerprint preparation but also mitigates model overfitting to improve efficiency further.

In the future, we plan to further validate the efficacy and robustness of WIFIND with additional datasets encompassing diverse indoor environments, including real-world scenarios with dynamic noise levels. Applying advanced positioning frameworks such as Transformer-based models or reinforcement learning approaches in WIFIND will be another potentially valuable direction to explore.

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